

EXERCISE 1

TUTORAT

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FL_02 - Les plateformes LLM de simulation sociale

1. Research focus

This document develops a literature review on LLM-based social simulation platforms and generative agent-based models. The objective is to understand how these systems attempt to address classical limitations of agent-based modelling (ABM) in social science—especially realism, calibration, and behavioral richness—while also examining the new methodological risks they introduce, including opacity, non-determinism, weak reproducibility, and governance challenges.

The review is deliberately positioned between two levels of analysis. First, it studies the main platforms, architectures, and evaluation strategies proposed between 2023 and 2025. Second, in light of the tutor’s feedback, it prepares a more specific direction for Exercises 2 and 3 by considering how a focused case study could be used to test the broader debate on “agentic” systems. A particularly relevant candidate is the OpenClaw/Moltbook ecosystem, which can be treated as a forward-looking bridge between designed social simulation platforms and more emergent agent-only environments.

2. Definitions

Classical ABM: Simulation with autonomous agents driven by explicit rules or heuristics; emergent macro outcomes arise from micro interactions.

Generative ABM / LLM-based social simulation: ABM where some decision components are replaced or augmented by an LLM, including memory, reflection, planning, dialogue, or action selection.

Platform (in this review): A reusable stack providing agent architecture, environment and orchestration, scalability support, logging or observability, and ideally evaluation hooks.

3. Research questions

RQ1. What are the classical bottlenecks of ABM in social simulation, especially realism, calibration, scalability, and external validity?

RQ2. What capabilities do LLM-driven agents add, such as language-native interaction, richer cognition, memory, and flexible planning?

RQ3. Do LLM-based ABMs solve classical ABM problems or shift them toward new issues such as bias, cost, non-determinism, and weak reproducibility?

RQ4. What are the main platforms and system designs in the recent literature, and how do they compare in architecture, scale, and evaluation claims?

RQ5. What evaluation and benchmarking practices are needed for trust, falsifiability, governance, and scientific usefulness?

RQ6. Under what conditions should emergent agentic ecosystems such as OpenClaw/Moltbook be treated as valid case studies for social simulation in Exercises 2 and 3?

4. Scope and boundaries

Time window: 2023–2025 for the core corpus, corresponding to the rise of generative agents and the first wave of LLM-based social simulation platforms.

Included: (i) platform and framework papers for social or societal simulation with LLM-driven agents; (ii) empirical studies that simulate specific social phenomena such as opinion dynamics, social movements, or social networks; and (iii) surveys and critical reviews that provide taxonomies, methodological discussion, or evaluation frameworks.

Excluded: generic LLM multi-agent orchestration papers that are not clearly connected to social simulation, unless they introduce concepts directly relevant to platform design, validation, or governance.

Forward-looking bridge: selected 2026 OpenClaw/Moltbook papers are not part of the core ranked corpus for Exercise 1, but they are retained as a possible case-specific direction for Exercises 2 and 3 because they speak directly to the ongoing debate on agentic social systems.

5. Literature identification methodology

Method: (A) seed paper approach based on course-relevant references and the existing draft; (B) backward search through reference lists to identify foundational architecture papers; (C) forward search to locate newer platforms, large-scale systems, and validation-oriented studies; and (D) targeted inclusion of high-quality surveys and critical reviews to structure concepts and evaluation dimensions.

Selection logic: priority was given to papers that are both recent and directly relevant to LLM-based social simulation platforms. Compared with the initial draft, the revised corpus reduces the weight of papers that are mainly about generic teaming or broad multi-agent systems, and strengthens the empirical social-simulation component with work on opinion dynamics, social movement simulation, social network simulation, and validation against real individuals.

6. Ranked reading list

The ranked reading list below follows the same documentary model as the initial draft—**Rank / Paper / Year / Venue / Citations (source) / Link**—but updates the corpus to better reflect the most relevant literature on LLM-based social simulation platforms and empirical social simulation studies.

Rank	Paper	Year	Venue	Citations (source)	Link
1	GenSim: A General Social Simulation Platform with Large Language Model based Agents	2025	NAACL 2025 System Demonstrations / ACL Anthology	41 (ACL Anthology)	Open link
2	AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society	2025	arXiv preprint	81 (Semantic Scholar snapshot)	Open link
3	Generative Agents: Interactive Simulacra of Human Behavior	2023	UIST 2023 / ACM Digital Library	4,500 (ACM Digital Library)	Open link

Rank	Paper	Year	Venue	Citations (source)	Link
4	S3: Social-network Simulation System with Large Language Model-Empowered Agents	2023	SSRN preprint / arXiv preprint	161 (SSRN)	Open link
5	Do Large Language Models Solve the Problems of Agent-Based Modeling? A Critical Review of Generative Social Simulations	2025	arXiv preprint	21 (Semantic Scholar snapshot)	Open link
6	Simulating Opinion Dynamics with Networks of LLM-based Agents	2024	Findings of NAACL 2024 / ACL Anthology	169 (ACL Anthology)	Open link
7	Unveiling the Truth and Facilitating Change: Towards Agent-based Large-scale Social Movement Simulation	2024	Findings of ACL 2024 / ACL Anthology	93 (ACL Anthology)	Open link
8	Large language models empowered agent-based modelling and simulation: a survey and perspectives	2024	Humanities & Social Sciences Communications (Nature Portfolio)	311 (Nature page snapshot)	Open link
9	From Individual to Society: A Survey on Social Simulation Driven by Large Language Model-based Agents	2024	arXiv preprint	62 (indexing snapshot)	Open link
10	GA-S3: Comprehensive Social Network Simulation with Group Agents	2025	Findings of ACL 2025 / ACL Anthology	9 (ACL Anthology)	Open link
11	Generative Agent Simulations of 1,000 People	2024	arXiv preprint	265 (Semantic Scholar snapshot)	Open link
12	LLMs and generative agent-based models for complex systems research	2024	Physics of Life Reviews / ScienceDirect	97 (ScienceDirect)	Open link

Citation counts are snapshots taken from the corresponding hosting or indexing pages in March 2026; they may evolve, especially for recent preprints.

6.1 Detailed entries

Detailed entries are abstract-focused and are included to preserve the structure of the original document while expanding the corpus from 10 to 12 papers.

1. GenSim: A General Social Simulation Platform with Large Language Model based Agents

Authors / Year: Tang et al., 2025

Venue: NAACL 2025 System Demonstrations / ACL Anthology

Link: <https://aclanthology.org/2025.naacl-demo.15/>

Citation count: 41 (ACL Anthology)

Abstract (summary): Proposes GenSim, an LLM-agent-based social simulation platform that abstracts reusable functions for customized scenario simulation, supports population-scale simulations (reported up to 100,000 agents), and incorporates correction mechanisms to improve reliability over long runs.

Why this paper is interesting: It is the most directly aligned platform paper for the tutorial topic because it frames itself as a general, reusable, and large-scale social simulation platform rather than a one-off demonstrator.

How I found it: Seeded by targeted keyword search on “social simulation platform” and validated through ACL Anthology.

2. AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances

Understanding of Human Behaviors and Society

Authors / Year: Piao et al., 2025

Venue: arXiv preprint

Link: <https://arxiv.org/abs/2502.08691>

Citation count: 81 (Semantic Scholar snapshot)

Abstract (summary): Introduces a large-scale social simulator that integrates LLM-driven agents, a realistic societal environment, and a large simulation engine. The paper reports social lives for over 10,000 agents and applications to polarization, inflammatory messaging, universal basic income, and external shocks.

Why this paper is interesting: It is one of the strongest large-scale platform papers because it connects platform design with concrete social-science use cases and explicit claims about alignment with real-world observations.

How I found it: Located through forward search from platform-focused keywords and confirmed on arXiv.

3. Generative Agents: Interactive Simulacra of Human Behavior

Authors / Year: Park et al., 2023

Venue: UIST 2023 / ACM Digital Library

Link: <https://dl.acm.org/doi/10.1145/3586183.3606763>

Citation count: 4,500 (ACM Digital Library)

Abstract (summary): Introduces generative agents powered by an LLM with natural-language memory, reflection, and planning. The paper demonstrates believable multi-agent behavior in a sandbox town, including coordination, routines, and emergent social activities.

Why this paper is interesting: This is the foundational architecture paper for the entire field. Many later social simulation platforms inherit or adapt its memory–reflection–planning logic.

How I found it: Identified through backward search because it is repeatedly cited as the foundational work in later platform and survey papers.

4. S3: Social-network Simulation System with Large Language Model-Empowered Agents

Authors / Year: Gao et al., 2023

Venue: SSRN preprint / arXiv preprint

Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5604712

Citation count: 161 (SSRN)

Abstract (summary): Presents S3, a social-network simulation system that uses the sensing, reasoning, and acting capacities of LLM agents to simulate user behavior, information diffusion, and interaction dynamics on social platforms.

Why this paper is interesting: It is one of the earliest papers to explicitly move from generic agents toward a social-network simulation system, making it important for the historical trajectory of the literature.

How I found it: Retrieved through targeted search combining “social-network simulation” and “LLM agents”, then cross-checked on SSRN and arXiv.

5. Do Large Language Models Solve the Problems of Agent-Based Modeling? A Critical Review of Generative Social Simulations

Authors / Year: Larooij & Törnberg, 2025

Venue: arXiv preprint

Link: <https://arxiv.org/abs/2504.03274>

Citation count: 21 (Semantic Scholar snapshot)

Abstract (summary): Critically reviews generative social simulations and argues that many strong claims are not yet supported by robust validation. The paper emphasizes risks such as opacity, weak causal explanation, reproducibility problems, and bias amplification.

Why this paper is interesting: It provides the main critical lens for the review and is essential for assessing whether LLM-based social simulations actually solve classical ABM limitations or merely shift them.

How I found it: Included as a key course-relevant critical paper and validated via arXiv and Semantic Scholar.

6. Simulating Opinion Dynamics with Networks of LLM-based Agents

Authors / Year: Chuang et al., 2024

Venue: Findings of NAACL 2024 / ACL Anthology

Link: <https://aclanthology.org/2024.findings-naacl.211/>

Citation count: 169 (ACL Anthology)

Abstract (summary): Studies opinion dynamics with populations of LLM-based agents interacting over networks. The paper shows the promise of richer behavior than classical ABMs, but also uncovers a strong model bias toward accurate information, which affects realism.

Why this paper is interesting: It is one of the strongest empirical papers in the corpus because it addresses a concrete social phenomenon instead of only describing a generic platform.

How I found it: Found through targeted searches on opinion dynamics and verified through ACL Anthology.

7. Unveiling the Truth and Facilitating Change: Towards Agent-based Large-scale Social Movement Simulation

Authors / Year: Mou, Wei, & Huang, 2024

Venue: Findings of ACL 2024 / ACL Anthology

Link: <https://aclanthology.org/2024.findings-acl.285/>

Citation count: 93 (ACL Anthology)

Abstract (summary): Develops an agent-based large-scale social movement simulation framework to study collective action, protest-like dynamics, and change processes in an online environment using LLM-driven agents.

Why this paper is interesting: It extends the literature from general platforms toward a socially meaningful application domain and helps show how LLM-agent simulations can be used for movement and diffusion studies.

How I found it: Identified through ACL Anthology searches on social movement simulation and checked against recent surveys.

8. Large language models empowered agent-based modelling and simulation: a survey and perspectives

Authors / Year: Gao et al., 2024

Venue: Humanities & Social Sciences Communications (Nature Portfolio)

Link: <https://www.nature.com/articles/s41599-024-03611-3>

Citation count: 311 (Nature page snapshot)

Abstract (summary): Surveys how LLMs are integrated into agent-based modeling and simulation across physical, cyber, and social domains. It discusses architectures, opportunities, evaluation, and methodological constraints.

Why this paper is interesting: This is one of the best high-visibility background surveys because it bridges traditional ABM and LLM-agent systems and offers a strong state-of-the-art overview.

How I found it: Retrieved through broad search on LLM-empowered ABM surveys and selected for venue quality and direct ABM relevance.

9. From Individual to Society: A Survey on Social Simulation Driven by Large Language Model-based Agents

Authors / Year: Mou et al., 2024

Venue: arXiv preprint

Link: <https://arxiv.org/abs/2412.03563>

Citation count: 62 (indexing snapshot)

Abstract (summary): Reviews simulation work driven by LLM-based agents and organizes the literature into three types: individual simulation, scenario simulation, and society simulation. It also summarizes architectures, objectives, evaluation methods, datasets, and benchmarks.

Why this paper is interesting: It is one of the most topic-aligned surveys for the tutorial because it focuses explicitly on social simulation and provides a useful progression from individuals to society-scale systems.

How I found it: Found through survey-focused queries on social simulation and retained because of its explicit “individual to society” framing.

10. GA-S3: Comprehensive Social Network Simulation with Group Agents

Authors / Year: Zhang et al., 2025

Venue: Findings of ACL 2025 / ACL Anthology

Link: <https://aclanthology.org/2025.findings-acl.468/>

Citation count: 9 (ACL Anthology)

Abstract (summary): Presents GA-S3, a comprehensive social network simulation system that introduces group agents to model populations with shared behavioral profiles, thereby improving scalability while preserving socially meaningful interaction patterns.

Why this paper is interesting: It is valuable because it moves beyond individual-only agent design and proposes a group-agent abstraction for large-scale network phenomena.

How I found it: Located through ACL Anthology searches for 2025 social simulation papers and selected as a direct continuation of S3-style work.

11. Generative Agent Simulations of 1,000 People

Authors / Year: Park et al., 2024

Venue: arXiv preprint

Link: <https://arxiv.org/abs/2411.10109>

Citation count: 265 (Semantic Scholar snapshot)

Abstract (summary): Introduces an agent architecture built from qualitative interviews with 1,052 real individuals, then evaluates how accurately the resulting agents reproduce survey responses, personality traits, and outcomes in experimental replications.

Why this paper is interesting: It is central for the validity discussion because it offers one of the strongest recent attempts to benchmark LLM-based behavioral simulation against real individuals.

How I found it: Found through forward search from Generative Agents and selected for its unusually strong empirical validation angle.

12. LLMs and generative agent-based models for complex systems research

Authors / Year: Lu et al., 2024

Venue: Physics of Life Reviews / ScienceDirect

Link: <https://www.sciencedirect.com/science/article/pii/S1571064524001386>

Citation count: 97 (ScienceDirect)

Abstract (summary): Reviews LLMs and generative ABMs in the context of complexity science, with particular attention to human behavior, social dynamics, and the methodological opportunities and pitfalls of this emerging research direction.

Why this paper is interesting: It broadens the review beyond purely social-simulation venues and helps position generative ABMs within a wider complex-systems perspective.

How I found it: Retrieved through review-oriented searches on generative ABMs and complex systems research.

7. Why these papers belong to the same literature

All selected papers focus on the same research object: LLM-driven agents interacting in an environment to simulate socially meaningful behavior at individual, group, network, or society levels. They differ in emphasis—some are foundational, some are platform-centric, some are empirical, and some are critical—but they converge on a common set of questions: how agents are designed, how environments are structured, how interaction scales, how realism is evaluated, and how methodological risks are handled.

Taken together, the corpus allows the review to move from foundational design (Generative Agents), to platform implementation (GenSim, AgentSociety, S3, GA-S3), to concrete social phenomena (opinion dynamics and social movements), and finally to methodological and epistemic critique (Larooij & Törnberg; Gao et al.; Lu et al.; Mou et al.). This makes the literature internally coherent and suitable for a concept-centric synthesis rather than a purely descriptive bibliography.

8. Best journal and best author (within this set)

Within the selected corpus, the most field-aligned high-visibility venue is Humanities & Social Sciences Communications because it directly connects LLM-based ABM work with broader social-science methodology. A broader but still very strong review venue is Physics of Life Reviews, which situates generative ABMs within complex systems research.

The strongest anchor author in this set is Joon Sung Park. Generative Agents is the foundational architecture paper for the field, and Generative Agent Simulations of 1,000 People extends that line toward stronger empirical validation. Together, these two papers make Park the clearest reference point across platform design, behavioral realism, and evaluation.

9. Concept-centric synthesis plan and concept matrix

To follow a concept-centric review structure, the synthesis should be organized by key concepts rather than by isolated paper summaries. The matrix below indicates which papers contribute most clearly to each concept bucket.

Concept buckets:

- A. Agent architecture — memory, reflection, planning, persona, decision loop
- B. Environment & interaction design — world model, events, institutions, social networks
- C. Scalability & systems engineering — orchestration, throughput, long-run stability, cost
- D. Validation & evaluation — micro realism, macro alignment, calibration, benchmarks

E. Interpretability & causal explanation — mechanistic insight, causal claims, transparency

F. Governance, ethics & reproducibility — bias, safety, non-determinism, replication

G. Applications — policy experiments, polarization, diffusion, teaming, movement dynamics

Paper	A	B	C	D	E	F	G
P1 GenSim	X	X	X	X		X	X
P2 AgentSociety	X	X	X	X		X	X
P3 Generative Agents	X	X		X			
P4 S3	X	X	X	X		X	X
P5 Critical Review				X	X	X	
P6 Opinion Dynamics	X	X		X		X	X
P7 Social Movement	X	X	X	X		X	X
P8 Nature Survey	X	X	X	X	X	X	
P9 Individual to Society	X	X	X	X		X	X
P10 GA-S3	X	X	X	X		X	X
P11 1,000 People	X			X	X	X	X
P12 Complex Systems Review				X	X	X	

Paper labels: P1 GenSim; P2 AgentSociety; P3 Generative Agents; P4 S3; P5 Critical Review; P6 Opinion Dynamics; P7 Social Movement Simulation; P8 Nature Survey; P9 From Individual to Society; P10 GA-S3; P11 1,000 People; P12 Complex Systems Review.

Planned writing flow: (1) introduce ABM and its classical limitations; (2) explain how LLM-agent architectures modify the ABM pipeline; (3) compare platforms by architecture, scale, and evaluation; (4) synthesize empirical social-simulation studies on opinion dynamics, social movements, and social networks; (5) discuss validation, causality, and governance; and (6) open toward a case-specific analysis for Exercises 2 and 3, potentially through the OpenClaw/Moltbook debate on emergent agentic social systems.